1. **Scope:**
   - **1.1** Title: OA-9243A(V)3/SR Antenna Tilting Group (ATG) Refurbishment
   - **1.2** Expected Period of Performance: TBD

2. **Reference:**
   - **2.1** EE110-LB-OMI-010 (REV. 2) Operation and Maintenance Instructions, Organizational, Intermediate, And Depot Level for Electrically Operated Units of Antenna Tilting Group OA-9243A/B(V)/SR (Lightweight);
   - **2.2** NAVSEA STANDARD ITEM (NSI) 009-04 (FY24) Quality Management System; provide
   - **2.3** NAVSEA STANDARD ITEM (NSI) 009-32 (FY24) Cleaning and Painting Requirements; accomplish

NOTE: All requirements stated are applicable to each unit provided by the contractor.

3. **Requirements**
   - 3.1 **General Specifications:**
     - **3.1.1** Quantity of two (QTY=2) OA-9243A(V)3/SR Antenna Tilting Group (ATG).
   - 3.2 **Equipment Requirements:**
     - **3.2.1** Each OA-9243A(V)3/SR Antenna Tilting Group (ATG) shall be restored in accordance with NAVSEA Technical Manual EE110-LB-OMI-010 Electrically Operated Units of Antenna Tilting Group OA-9243A(V)3/SR (Lightweight) and OEM Technical Specifications.
   - 3.3 **Contractor Requirements:**
     - **3.3.1** Contractor shall provide complete overhaul, upgrades, and testing of OA-9243A(V)3/SR Antenna Tilting Group (ATG); completely dissemble ATG, perform repairs, paint/upgrade to powder coating components, provide upgraded parts for brakes, limit switches, gear box, bumper stops, and junction box as applicable. All powder coating / paint operations shall be in accordance with (NSI) 009-32 (2.3). Reassemble, perform alignments/adjustments, and operational testing to include:
       - **3.3.1.1** Motor: Remove Dings brake from motor, sand blast motor, send motor to a NAVSEA approved source to be completely checked out (at a minimum, replace bearings and seals), use 316 stainless steel hardware to assemble motor cases, paint motor with Ameron PSX700, color 26270, powder coat all Dings brake parts, and upgrade motor brake assembly by installing a corrosion resistant brake hub kit, L-3 P/N: 11910.
       - **3.3.1.2** Gearbox: Completely disassemble gearbox, replace bearings and retaining rings as required, sandblast external gearbox sections, upgrade gearbox, P/N: 0003-4010-401, output housing by making it from 316 stainless steel, powder coat gearbox sections, and assemble gearbox using new seals and grade 316 stainless steel hardware.
       - **3.3.1.3** Handcrank assembly: Completely disassemble handcrank, replace all bearings, thrust washers, and seals, powder coat external surfaces as required, and assemble using all 316 stainless steel hardware.

3.3.1.4 Limit Switch Assembly: Replace old limit switch assemblies with upgraded limit switch assemblies, L-3 P/N: 12221-1, strip limit switch cams and overtravel limit switch cams of all paint, passivate stainless steel parts, powder coat all cam plates, and assemble cam plate assemblies with 316 stainless steel hardware.

3.3.1.5 Pedestal: Remove all ground bosses, install new ground bosses per MIL-STD-1310H using the three strap configuration, sandblast pedestal to a near white condition, and powder coat pedestal.

3.3.1.6 Mast: Remove all ground bosses, install new ground bosses per MIL-STD-1310H using the three strap configuration, mask all surfaces and threaded holes which do not require powder coating, inspect mast for damage and repair as required, sandblast to a near white condition, and powder coat mast.

3.3.1.7 Antenna Support Arm: Mask all threaded holes and sandblast to a near white condition and powder coat antenna support arm.

3.3.1.8 Sub Assembly Gearbox and Motor to Pedestal: install gearbox into pedestal using new grade 316 ½-20UNF HEX-HD cap screws, coat HEX-HD CAP screws with anti-seize per MIL-T-22361, and install motor onto gearbox using grade 316 stainless steel hardware.

3.3.1.9 Main Junction Box: Replace old junction box assembly with an upgraded main junction box, L-3 P/N: 13524, replace old selector switch with L-3 P/N: 12223 antenna products selector switch kit, (if required) replace wiring harness, replace main junction box cover gasket and hardware, rebuild overtravel limit switch assembly and overtravel limit switch plunger assembly, install main junction box and upgraded limit switch assembly onto gearbox and pedestal using 316 stainless steel hardware.

3.3.1.10 Mast Installation to Gearbox: Install mast using 316 stainless steel hardware, install handcrank assemblies, and fill gearbox with 75W gearbox oil.

3.3.1.11 Installation of Antenna Support Arm: Install antenna support arm using new 5/8-18UNF-2A, grade 316 stainless steel HEX-HD cap screws and install cover plate using same hardware as antenna support arm.

3.3.1.12 Adjustment of Limit Switch Cams and Overtravel Limit Switch Cams: Using a precision level, set ATG to its vertical position, set overtravel limit switch upper limit cam to – 1° of the limit switch cam, using a precision level, handcrank the ATG to its horizontal position, adjust cam, and set overtravel limit switch cam to +1° of the limit switch cam.

3.3.1.13 Sandblast and powder coat steel coupler mount, or install New Stainless Steel Coupler Mount, L-3 P/N: 12823-1

3.3.1.14 Label Plates: Install new safety, caution, and ID plates.

3.3.1.15 Final, Post-Overhaul Acceptance Testing **(I) (G) INSPECTION,** (NSI) 009-04 (2.2) refers: Contractor shall host NNSY to witness final, post-overhaul acceptance testing of refurbished OA-9243A(V)3/SR Antenna Tilting Groups. This shall be done before ATGs depart from overhauling facility to be returned to NNSY as "A"-condition assets to ensure satisfactory ATG operation and verify Contractor quality of work.

<u>Government Requirements:</u>

3.3.1 Provide two EA F-Condition OA-9243A(V)3/SR Antenna Tilting Group (ATG) "cores" for refurbishment.

3.3.2 Provide technical representative(s) from NNSY to witness final, post-overhaul acceptance testing of refurbished OA-9243A(V)3/SR Antenna Tilting Groups before ATGs depart from overhauling facility to be returned to NNSY as "A"-condition assets.

4. Points Of Contact:

4.1 ***Technical POC***: TBD

4.2 ***Contracting Officer's Representative (COR):*** TBD